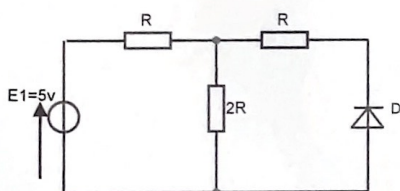


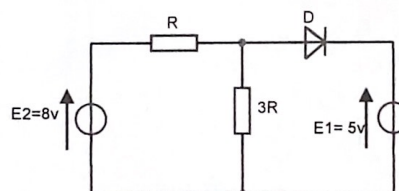
MONTAGES AVEC DIODE

- ☐ Modèle d'un interrupteur (diode) et d'une source
- ☐ Révisions : loi des nœuds et des mailles

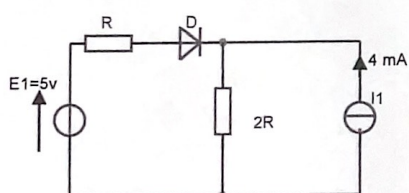
Pour les différents montages ci-dessous, déterminer l'état de la diode et la valeur du courant qui la traverse, ou de la tension à ses bornes.

 $R = 1\text{k}\Omega$


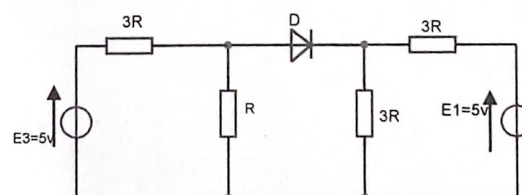
Montage 1



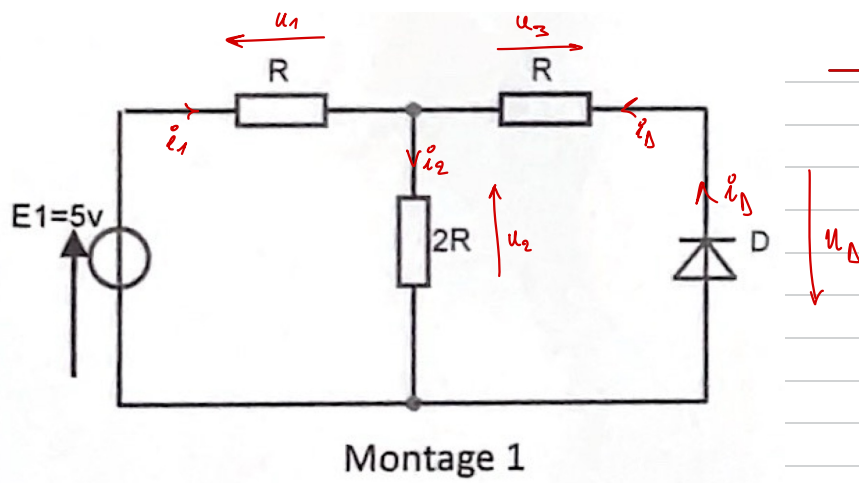
Montage 2



Montage 3



Montage 4



$$i_1 + i_D = i_2 \quad \text{LKN}$$

$$\left. \begin{aligned} u_2 + u_1 &= E_1 \\ u_2 &= -u_3 - u_D \end{aligned} \right\} \text{LKM}$$

$$\left. \begin{aligned} 2R i_2 + R i_1 &= E_1 \\ 2R i_2 &= -R i_D - u_D \end{aligned} \right\} \text{loi d'Ohm}$$

$$\rightarrow 2R i_2 + R (i_2 - i_D) = E_1$$

$$\rightarrow 3R i_2 - R i_D = E_1 \rightarrow 3R i_2 = E_1 + R i_D$$

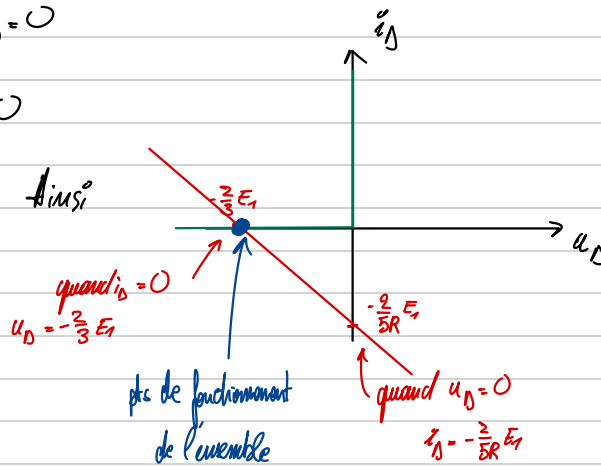
$$\rightarrow \frac{2}{3} (E_1 + R i_D) = -R i_D - u_D$$

$$\rightarrow \frac{2}{3} E_1 + \frac{2}{3} R i_D + R i_D + u_D = 0$$

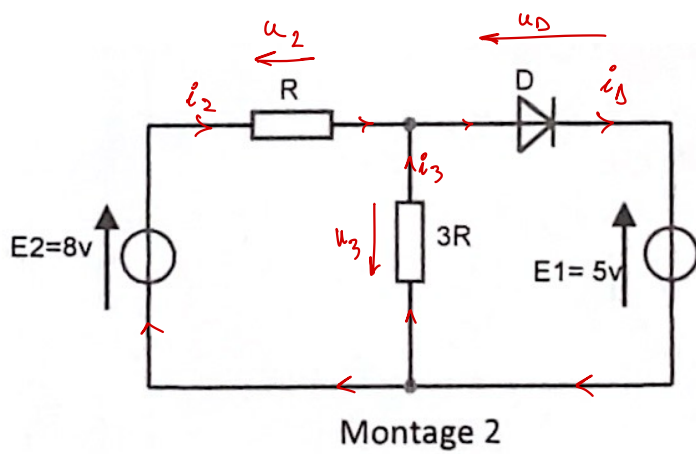
$$\rightarrow \frac{2}{3} E_1 + \frac{5}{3} R i_D + u_D = 0$$

$$u_D = -\frac{2}{3} E_1 - \frac{5}{3} R i_D$$

Ainsi



$i_D = 0$ dans diode bloquée



$$i_D = i_2 + i_3 \quad \left. \vphantom{i_D} \right\} \text{LDN}$$

$$\begin{aligned} u_2 &= E_2 + u_3 \\ E_1 &= -u_D - u_3 \end{aligned} \quad \left. \vphantom{u_2} \right\} \text{LDN}$$

$$\begin{aligned} R i_2 &= E_2 + 3R i_3 \\ E_1 &= -u_D - 3R i_3 \end{aligned} \quad \left. \vphantom{R i_2} \right\} \text{Ohm}$$

$$\Rightarrow R i_2 = E_2 + 3R (i_D - i_2)$$

$$R i_2 = E_2 + 3R i_D - 3R i_2$$

$$4R i_2 = E_2 + 3R i_D$$

$$\Rightarrow E_1 = -u_D - 3R (i_D - i_2)$$

$$E_1 = -u_D - 3R i_D + 3R i_2$$

$$3R i_2 = E_1 + u_D + 3R i_D$$

$$R i_2 = \frac{1}{3} E_1 + \frac{1}{3} u_D + R i_D$$

$$4 \left(\frac{1}{3} E_1 + \frac{1}{3} u_D + R i_D \right) = E_2 + 3R i_D$$

$$\frac{4}{3} E_1 + \frac{4}{3} u_D + 4R i_D = E_2 + 3R i_D$$

$$\frac{4}{3} E_1 + \frac{4}{3} u_D + R i_D = E_2$$

$$\frac{4}{3} u_D = E_2 - \frac{4}{3} E_1 - R i_D$$

$$u_D = \frac{1}{4/3} E_2 - \frac{4/3}{4/3} E_1 - \frac{1}{4/3} R i_D$$

$$u_D = \frac{3}{4} E_2 - E_1 - \frac{3}{4} R i_D$$

$$\text{AN: } u_D = \frac{3}{4} \times 8 - 5$$

$$= 3 \times 2 - 5$$

$$\boxed{u_D = 1}$$

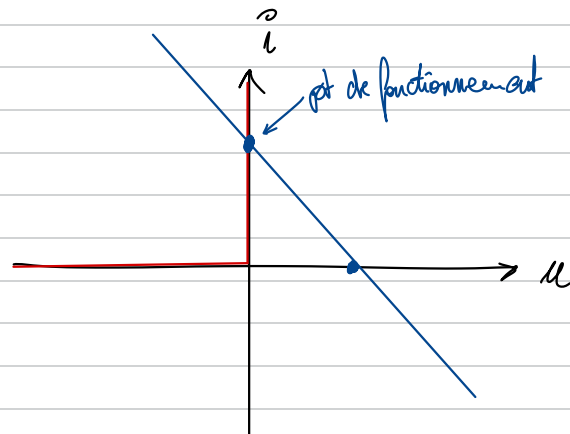
$$\boxed{u_D = \frac{3}{4} E_2 - E_1}$$

$$\text{AN: } i_D = \frac{8}{R} - \frac{4}{3} \frac{E_1}{R} > 0$$

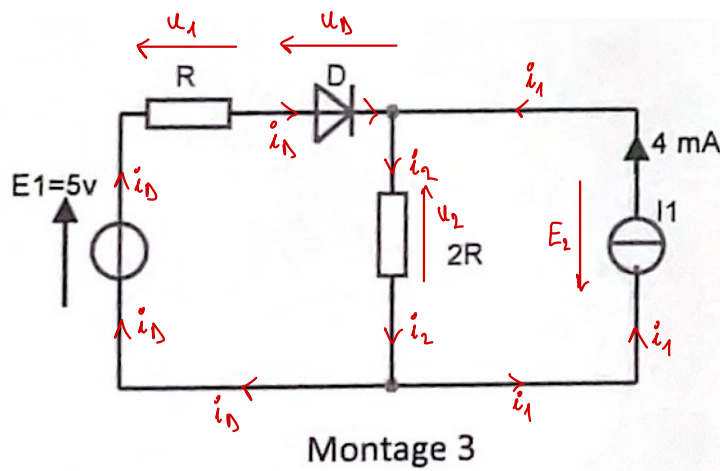
$$\frac{3}{4} R i_D = \frac{3}{4} E_2 - E_1$$

$$\frac{3}{4} i_D = \frac{3}{4} \frac{E_2}{R} - \frac{E_1}{R}$$

$$\boxed{i_D = \frac{E_2}{R} - \frac{4}{3} \frac{E_1}{R}}$$



La diode est passante



$$i_2 = i_0 + i_1 \quad \left. \vphantom{i_2} \right\} \text{LDN}$$

$$\left. \begin{aligned} E_1 &= u_1 + u_D + u_2 \\ E_2 + u_2 &= 0 \end{aligned} \right\} \text{LDM}$$

$$\left. \begin{aligned} E_1 &= R i_D + u_D + 2R i_2 \\ E_2 + 2R i_2 &= 0 \end{aligned} \right\} \text{Ohm}$$

$$\Rightarrow 2R i_2 = -E_2$$

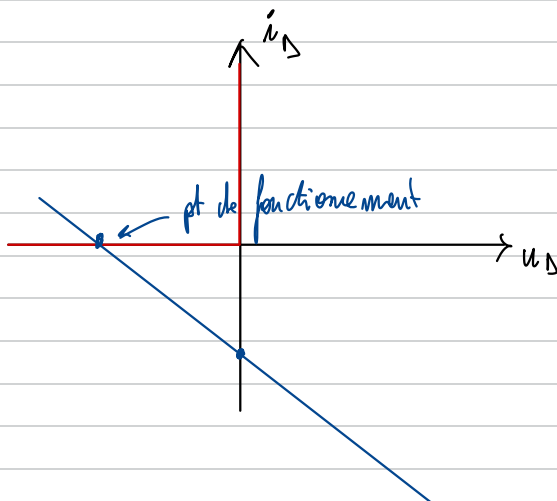
$$\Rightarrow E_1 = R i_D + u_D - E_2$$

$$\Rightarrow u_D = E_1 + E_2 + R i_D$$

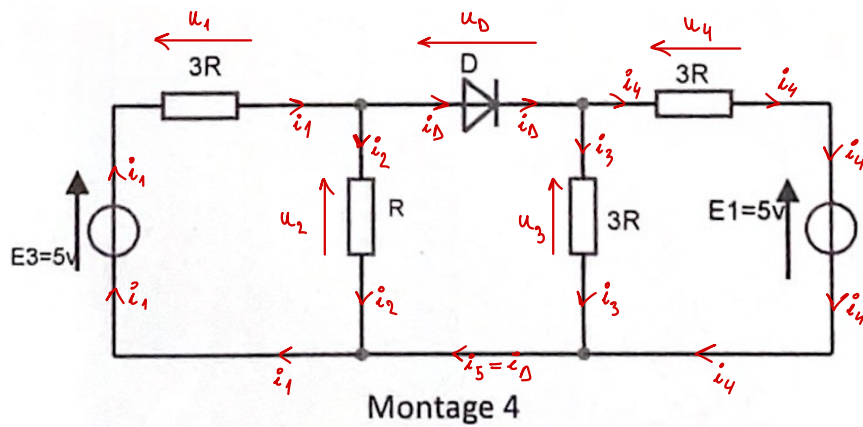
$$\Rightarrow i_D = \frac{E_1}{R} + \frac{E_2}{R} + \frac{1}{R} u_D$$

$$\text{quand } i_D = 0 \quad u_D = E_1 + E_2$$

$$\text{quand } u_D = 0 \quad i_D = \frac{E_1}{R} + \frac{E_2}{R}$$



Diode bloquée



LDN: $i_1 = i_2 + i_D$
 $i_D = i_3 + i_4$
 $i_5 = i_4 + i_3$
 $i_1 = i_2 + i_5$

Au final $i_5 = i_D \Rightarrow$ Equations utiles: $i_1 = i_2 + i_D$
 $i_D = i_3 + i_4$

LDM: $E_3 = u_1 + u_2$

$u_2 = u_D + u_3$

$u_3 = u_4 + E_1$

Ohm: $E_3 = 3Ri_1 + Ri_2 \quad (1)$

$Ri_2 = u_D + 3Ri_3 \quad (2)$ $i_1 = i_2 + i_D$
 $i_D = i_3 + i_4$

$3Ri_3 = 3Ri_4 + E_1 \quad (3)$

(2) $u_D = Ri_2 - 3Ri_3$ (1) $E_3 = 3Ri_1 + R(i_1 - i_D)$ (3) $3Ri_3 = 3Ri_4 + E_1$

$= R(i_1 - i_D) - 3R(i_D - i_4)$ $= 3Ri_1 + Ri_1 - Ri_D$ $3R(i_D - i_4) = 3Ri_4 + E_1$

$= Ri_1 - Ri_D - 3Ri_D + 3Ri_4$ $= 4Ri_1 - Ri_D$ $3Ri_D - 3Ri_4 = 3Ri_4 + E_1$

$= -4Ri_D + Ri_1 + 3Ri_4$ $4Ri_1 = E_3 + Ri_D$ $6Ri_4 = 3Ri_D - E_1$

$i_1 = \frac{1}{4} \frac{E_3}{R} + \frac{1}{4} i_D$ $i_4 = \frac{1}{2} i_D - \frac{1}{6} \frac{E_1}{R}$

$$u_D = -4Ri_D + R\left(\frac{1}{4}\frac{E_3}{R} + \frac{1}{4}i_D\right) + 3R\left(\frac{1}{2}i_D - \frac{1}{6}\frac{E_1}{R}\right)$$

$$= -4Ri_D + \frac{1}{4}E_3 + \frac{1}{4}Ri_D + \frac{3}{2}Ri_D - \frac{1}{2}E_1$$

$$u_D = -\frac{9}{4}Ri_D + \frac{1}{4}E_3 - \frac{1}{2}E_1 \quad \checkmark$$

Quand $i_D = 0 \Rightarrow u_D = \frac{1}{4}E_3 - \frac{1}{2}E_1$ AN: $\frac{1}{4} \times 5 - \frac{1}{2} \times 5 = -\frac{5}{4}$

Quand $u_D = 0 \Rightarrow \frac{9}{4}Ri_D = \frac{1}{4}E_3 - \frac{1}{2}E_1$

$$\Rightarrow i_D = \frac{1}{9}\frac{E_3}{R} - \frac{2}{9}\frac{E_1}{R} \quad \text{AN: } \frac{1}{9} \times \frac{5}{10^3} - \frac{2}{9} \times \frac{5}{10^3} = -\frac{1}{1800}$$

